mmFallBbox: An Effective and Efficient Fall Detection Framework With a Comprehensive Benchmark

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***Abstract*—Accidental falls among the elderly demand highly reliable detection systems; however, existing solutions based on wearables, cameras, or WiFi often suffer from environmental interference, privacy concerns, and poor performance in detecting slow-onset falls (e.g., fainting). Despite these challenges, millimeter-wave (mmWave) radar-based methods have emerged as an attractive alternative. In this article, we propose an effective and efficient fall detection method, mmFallBbox, which leverages mmWave radar to track 3-D human bounding box dynamics. By exploiting the correlation between fall states and bounding box evolution, our approach effectively distinguishes between normal activities and slow fall states, such as fainting or medical conditions, which traditional systems struggle to detect. To evaluate the performance of mmFallBbox, we collected a large-scale fall detection dataset consisting of 60 h of radar data synchronized with video annotations. This dataset will be made publicly available to the research community for further development. We achieved an $F1$ score of 0.977 on our dataset and achieved state-of-the-art performance with limited computational complexity. Moreover, extensive experimental results show significant improvements in detecting slow-onset falls and providing explainable outputs, offering a promising solution for real-world fall detection applications.**

***Index Terms*—Deep learning, fall detection, millimeter-wave (mmWave) radar.**

I. Introduction

ACCIDENTAL falls constitute a major safety hazard for older adults and have attracted increasing research attention. Although wearable-device-based fall detection systems exist, they often cause user discomfort, which limits their adoption in practice. Recent advances in computer vision have enabled considerable progress in vision-based fall detection [1], [2]. Nevertheless, camera-centric approaches still face several obstacles. First, cameras perform poorly in environments with occlusions or insufficient illumination. Second, the continuous video monitoring they require raises privacy concerns that impede large-scale deployment in smart-home and healthcare settings.

Over the past few years, wireless-signal-based fall detection schemes have garnered substantial research interest [3], [4]. Representative techniques exploit channel state information (CSI) extracted from commodity Wi-Fi or frequency-modulated continuous-wave (FMCW) radar to analyze radio reflections from the human body. A fall induces rapid acceleration and displacement, which appear as pronounced fluctuations in the measured CSI features [3], [4], [5]. Nevertheless, CSI-driven approaches often generalize poorly across diverse settings and are susceptible to false alarms triggered by other forms of high-speed human motion.

FMCW radar offers high resolution in both velocity and distance. Existing methods detect fall events by performing time–frequency analysis on radar signals and extracting features such as signal peaks and power burst curves [6], [7]. Nevertheless, these approaches demonstrate limited generalization across different environments and are susceptible to false positives triggered by other rapid motions. When combined with multiple-input and multiple-output (MIMO) antennas, FMCW radar provides angular resolution. Given the inherent difficulty in interpreting radar signals, existing methods employ neural networks (NNs) to extract complex features for improved fall detection accuracy [8], [9], [10], [11], [12], [13]. For instance, Tian et al. [8] utilized heatmaps from two orthogonally positioned radars and a 3D-ResNet to extract spatiotemporal features. Jin et al. [14] assumed that radar point clouds follow a specific multivariate Gaussian distribution and learned this distribution for fall detection using variational inference.

The current dominant approach, which treats falls as isolated events within fixed-time sliding windows, is increasingly deemed inadequate. In real-life situations, individuals may experience scenarios where they collide with a wall or furniture, lose consciousness, and subsequently slump or lean against nearby objects. Similarly, users suffering from conditions such as a heart attack or hypotension may enter a confused or debilitated state, with a loss of muscle control. In such instances, falls often lack the distinct height or velocity signatures typically associated with a standard fall. Existing detection systems, almost universally, fail to identify these nuanced cases, even though it is crucial that such systems still trigger an alarm under these circumstances.

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Furthermore, although contemporary fall detection systems incorporate various filtering techniques and interframe coherent integration methods to mitigate multipath effects and improve generalization across different environments, the outcomes remain suboptimal. This is primarily because filtering can unintentionally eliminate useful signal components, leading to intrinsic architectural limitations that impede the system's ability to handle a wide range of boundary conditions.

Existing fall detection systems currently lack the necessary adaptations to handle exceptional or extreme scenarios effectively. Based on our testing, we observe that false positives and false negatives frequently occur under certain conditions.

False positives, where nonfall events are mistakenly classified as falls, are commonly triggered in the following situations.

1) Robotic vacuums or pets (e.g., cats or dogs) moving through the detection area.
2) Free-falling nonhuman objects.

False negatives, where actual falls are missed by the system, tend to happen in these cases.

1) Kicking a chair or table during the fall.
2) Waving arms for help, crawling, or rolling after the fall.
3) Rolling off a bed.
4) Interference from water flow during showering.
5) Nearby electric fans or thick, swaying curtains.

These situations complicate the detection process, as they often obscure key features that are typically associated with falls. Given the transient nature of these events, it becomes inherently difficult to accurately classify a brief radar signal segment (such as a three-second window) as "fall" or "nonfall" using traditional detection methods.

Recent studies have explored indirect fall detection using radar by monitoring body poses [15], [16], [17], [18], [19]. However, unlike cameras, which capture projections of all unoccluded body parts, mmWave radar only receives reflections from a limited subset of body parts in each snapshot [20], [21]. This results in inherently inaccurate pose estimation. Furthermore, for real-time fall detection systems, the sparse nature of this data leads to delayed radar responses. To address these issues, we propose simplifying the pose representation. Our objective is to derive continuous and interpretable data that can simultaneously meet the dual requirements of accuracy and real-time performance.

In this article, we introduce mmFallBbox, a system that indirectly detects falls by tracking the 3-D human bounding box. We leverage radar to extract spatiotemporal information and preprocess radar heatmaps using advanced localization algorithms. Although multipath propagation severely limits the cross-domain performance of RF signals, these multipath reflections inherently carry valuable information. To address this challenge, we propose a novel deep convolutional network-based location mapping (Loc-Map) model. This model effectively decouples target location information from features induced by multipath effects, allowing for the full utilization of available data while minimizing environmental interference. Importantly, in addition to providing a binary fall classification output, our model also predicts the dimensions (length, width, and height) of the detected bounding box. This dual output significantly enhances the system's explainability and provides the capability to issue early warnings for other anomalous activities.

Our main contributions are as follows.

1) We propose mmFallBbox, a mmWave radar system that tracks 3-D human bounding box dynamics to detect falls. By correlating changes in position and size with fall states, the system effectively identifies both rapid and slow-onset falls (e.g., fainting), which are often missed by motion- or height-based methods.
2) We introduce Loc-Map Net, a deep network that decouples target localization from multipath-induced features. It adaptively leverages multipath reflections while suppressing interference, greatly improving generalization and robustness across environments.
3) We have constructed a comprehensive RF-Vision fall dataset, comprising 60 h of data collected in both vertical and horizontal radar configurations. The dataset spans five distinct environments, ten different perspectives, and includes data from 50 diverse individuals. To simulate real-world corner cases, the dataset features challenging interference scenarios such as fluttering curtains, rotating fan blades, running water, and multiperson interactions.

II. Related Work

A. Fall Detection

Research on fall detection has been extensive [22]. Wearable devices typically employ sensors such as gyroscopes [23] and accelerometers [24], [25]. While simple and generally accurate, this sensor-based approach often relies on postfall factors like body angle and peak acceleration induced by impact with the ground. Lee and Tseng [25] utilized smartphone-embedded accelerometers, achieving a detection accuracy of 96%. Cai et al. [26] attained 99.08% accuracy using a waist-worn triaxial accelerometer.

Vision-based methods extract human pose information via NNs. Ramirez et al. [27] employed human skeleton estimation to extract features in their detection method. De Miguel et al. [28] combined background subtraction, Kalman filtering, and optical flow as input for a machine learning algorithm, achieving high detection accuracy. Shu and Shu [29] monitored the human 2-D bounding box using eight cameras to implement their system in practice.

CSI-based methods leverage Wi-Fi CSI [3], [4], employing short-time Fourier transform (STFT) or wavelet transforms to detect significant changes in frequency-domain signals. Doppler-based methods [6], [7], [30] utilize velocity variations in the time domain.

Millimeter-wave (mmWave) radar-based approaches are categorized into single-antenna and multiantenna methods. Single-antenna methods utilize the radar's velocity and ranging modules. Radar units are often mounted overhead to capture vertical, free-fall-like velocity changes and ground impact using the velocity module, or to detect low-height information indicating a person is on the ground using the ranging module [7], [31], [32].

Combining radar with MIMO technology provides angular resolution, offering higher-dimensional spatial information but also increasing signal complexity. Existing methods employ beamforming to achieve spatial resolution, generating point

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TABLE I
COMPARISON OF RF-BASED FALL DETECTION METHODS

Method	Hardware	Data Processing	Characteristics Selection	Detection Method
WiFall [4]	WiFi	Signal Amplitude Extraction	Amplitude feature	SVM
FallDefi [3]	WiFi	STFT	Spectrum feature	SVM
Sadreazami et al.[7]	Doppler Radar(single antenna)	STFT	Binary time-frequency signature	CNN
Saeed et al. [31]	Doppler Radar(single antenna)	STFT	Spectrum heatmap	CNN
LT-Fall [10]	MIMO mmWave Radar	4D Point Cloud Extraction	Height	Threshold
mmFall [14]	MIMO mmWave Radar	4D Point Cloud Extraction	Gaussian fitting	VAE
Ding et al. [35]	MIMO mmWave Radar	Range-Doppler Map (RDM)	-	Conv-LSTM
Aryokee[8]	MIMO mmWave Radar	Range-Angle Map (RAM)	-	Resnet
mmFallBbox(Ours)	MIMO mmWave Radar	Range-Angle Map (RAM)	3D Bounding Box	CNN + Threshold

clouds or heatmaps for visual spatial information. Fall detection is then performed based on point cloud features using thresholding. However, since radar often captures only a subset of reflections, further processing or NN-based feature extraction is required to fully utilize the point cloud or raw radar signals.

Zhang et al. [10] refined the fall identification criteria, arguing that only severe, life-threatening falls warrant alerts. They defined life-threatening falls as those involving a fall from height or resulting in lying on the floor, incorporating a state of stillness into their judgment logic to improve accuracy.

Table I presents the hardware, signal processing methods, characteristics selection, and detection methods of existing wireless-signal-based fall detection approaches.

B. NN Applications in Fall Detection

NNs, capable of extracting task-relevant features from complex information, have been widely applied in fall detection. They were first extensively utilized in vision-based fall detection methods [27], [33], achieving impressive performance. Wireless signals, containing echoes from reflecting objects, possess temporal and spatial resolution and record user motion information. Consequently, NNs can be employed to extract these intricate and often difficult-to-interpret spatiotemporal features.

Tian et al. [8] employed a two-branch NN to process vertical and horizontal heatmaps separately, followed by a cascaded NN for two-stage fall classification. Sun et al. [34] utilized range-angle heatmaps combined with a long short-term memory (LSTM) network. Ding et al. [35] performed a fast Fourier transform (FFT) on the fast-time domain within a sliding window over the slow-time domain to obtain dynamic range–Doppler frames (DRDFs), further enhancing performance over the method proposed by Tian et al. [8]. Yao et al. [36] proposed using three NN modules to extract features from a single range–Doppler map (RDM) and separate vertical and horizontal range-angle maps (RAMs), fusing spatial and velocity information before final classification.

However, NNs introduce new challenges in generalization. Saeed et al. [31] addressed this by expanding data collection across environments using micro-Doppler techniques. They collected data from five distinct environments without altering radar deployment configurations. Ding et al. [37] designed an NN to estimate scene flow as complementary motion information for mmWave radar point clouds, aiming to improve generalization for downstream tasks. Jin et al. [14] mitigated the impact of environment on point cloud estimation by fitting the point cloud to a 3-D normal distribution, effectively discarding most interference information. Wispeed [38] ingeniously leveraged multipath effects, utilizing the signal's autocorrelation function to estimate environmental signal variations for velocity estimation. This approach transformed multipath interference from a nuisance factor into an effective estimation parameter.

III. MMFALLBBOX SYSTEM

The mmFallBbox system consists of five modules, as shown in Fig. 1: it first acquires synchronized image and radar data. The ground-truth bounding box generation module processes the visual signals to obtain ground-truth bounding boxes of humans. the signal processing module processes the radar signals into interpretable heatmaps containing 3-D angle-of-arrival (AoA), time-of-flight (ToF), and radar cross section (RCS) information. To obtain target location information, the location module employs a path selection algorithm to estimate human positions and suppress interference from dynamic multipath reflections [39], obtaining target coordinates and cropping the localized regions. Subsequently, the full heatmap sequences and the cropped regions are fed into a NN to jointly predict bounding boxes and fall status. Finally, the predicted bounding boxes, along with a multithreshold sequence, are used to visually interpret the fall event.

A. Signal Processing

We utilize an FMCW radar with linear antenna arrays to receive reflections from different spatial locations. By combining FMCW and the antenna array, the RF signals from the location (x, y, z) can be extracted as

$$y(x, y, z, t) = \sum_{k=1}^{K} \sum_{m=1}^{M} s_{k,m,t} \cdot e^{j2\pi \frac{d_m(x,y,z)}{\lambda_k}} \tag{1}$$

where $s_{k,m,t}$ denotes the kth sample of the FMCW sweep on the mth antenna at time t, λ_k is the signal wavelength, and $d_m(x, y, z)$ denotes the round-trip distance from (x, y, z) to the radar. The computational complexity of processing a 4-D tensor is extremely high [40]. Hence, we employ only a horizontal antenna array to reduce hardware requirements and data complexity. The heatmaps can be given by

$$y_{hor}(x, y, t) = \sum_{k=1}^{K} \sum_{m=1}^{M} s_{k,m,t} \cdot e^{j2\pi \frac{d_m(x,y)}{\lambda_k}}. \tag{2}$$

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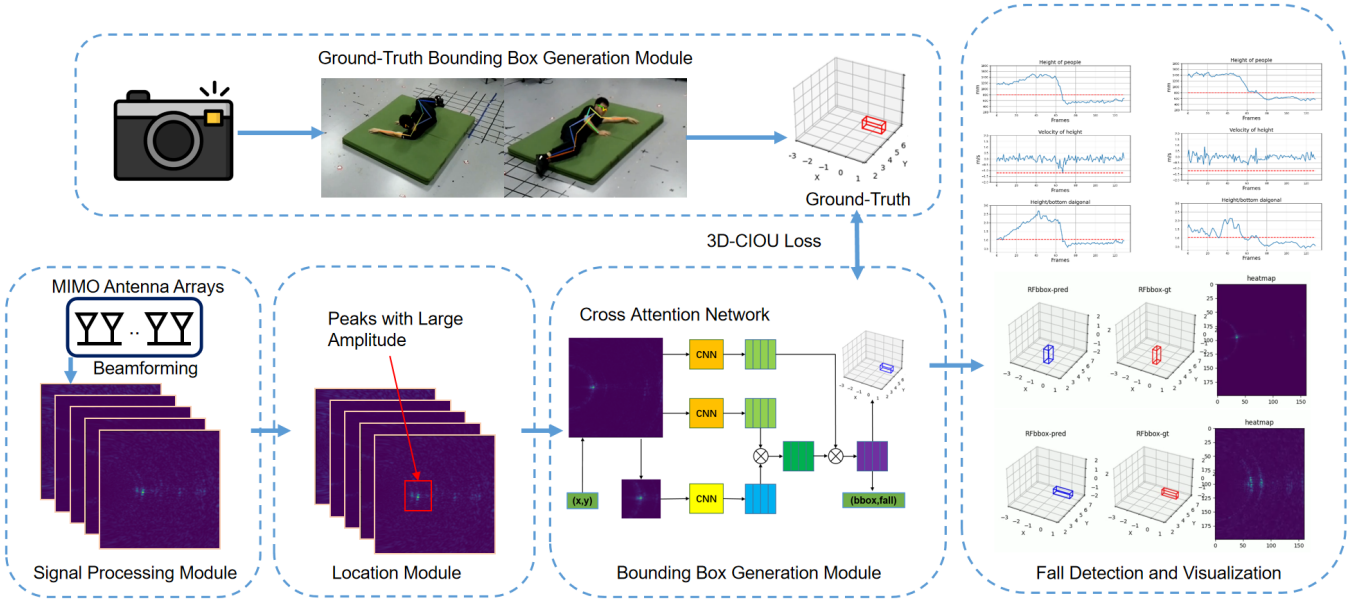


Fig. 1. Overview of mmFallBbox. mmFallBbox first acquires synchronized visual and radar signals. It processes the visual signals to obtain ground-truth bounding boxes of humans. For radar signals, it applies beamforming to generate a sequence of RAMs and employs a localization algorithm to obtain target coordinates and crop the localized regions. Both the full heatmap and the localized regions are fed into a cross-attention network for joint bounding box prediction. The system further enhances explainability through multithreshold decision logic and visualized bounding box outputs.

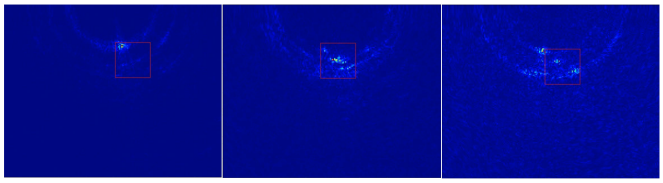


Fig. 2. From left to right are the RAMs for standing, sitting, and falling, respectively.

As shown in (2), we can obtain the (x, y) coordinates of the target in the heatmap, and the height information z of the target is also reflected in the signal strength of the heatmap. This is because when a fall occurs, the RCS in the vertical direction decreases, resulting in a weakening of the signal strength [41]. Therefore, we can use a single heatmap to generate bounding boxes and perform fall detection. The RCS is given by the following equation:

$$\sigma = \lim_{r\to\infty} 4\pi r^2 \frac{S_s}{S_i} \tag{3}$$

where r is the target distance, and S_s and S_i are the measured scattering and incident densities at the target, respectively. RCS is influenced by factors such as target material, target size, and incidence angle. For the human body, the material remains consistent, and the incidence angle falls within 120° directly in front of the radar. Moreover, signal processing via beamforming ensures a consistent incident energy density. When a person falls from a sitting posture, the signal strength in the vertical direction decreases, while the area covered by the human body expands. Fig. 2 illustrates the signal maps for standing, sitting, and falling postures. It can be observed that when standing, there is a distinct bright spot with low energy in the surrounding area. In the sitting posture, the energy of the human torso slightly attenuates, accompanied by high-energy regions nearby. During a fall, compared to standing and sitting, no significant bright spot appears within the human body area; instead, the high-energy area expands, and the background exhibits noise caused by multipath reflections from the fall event.

We apply differentiation to the time-domain signals to filter out interference from static environments.

B. Location

As derived from (2), the amplitude within the AoA–ToF map represents the reflected signal power from a specific point. Consequently, amplitude can indicate the likelihood of target presence. Simply put, the point exhibiting the maximum amplitude corresponds to the target's location. However, this approach is prone to misjudging targets due to multipath interference. Multipath interference is categorized into static multipath and dynamic multipath. Static multipath originates from reflections off static objects. Owing to factors like surface area and material properties, static multipath components are often very strong. Therefore, relying solely on amplitude for target localization tends to erroneously identify the positions of static objects. Fortunately, mitigating static multipath is relatively straightforward. Since static multipath components remain constant (static) over time, they can be effectively eliminated by applying signal differencing techniques. This process simultaneously preserves the signals from moving targets. Consequently, selecting the region with the maximum amplitude in the differenced signal yields coarse localization information.

However, the precision and robustness of localization remain constrained due to the presence of dynamic multipath. Dynamic multipath arises because a moving person not only reflects signals directly but also alters reflection paths from other objects. For instance, signals may reflect from the human

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body to a wall before being received, or reflect from a wall to the human body and then to the receiver, creating ghost targets. These multipath signals are similar to the signals from the moving person and exhibit temporal variability, rendering them resistant to simple differencing techniques. To address this challenge, we exploit a key distinction: the position of a genuine moving target evolves continuously, whereas the apparent positions of dynamic multipath artifacts exhibit rapid, random fluctuations. This characteristic enables the filtering of points exhibiting abrupt signal changes. Specifically, we define a cost function characterizing signal variation over a fixed time period

$$c_i = -\left|\boldsymbol{M}'_{ai}\right| - \left|\boldsymbol{M}'_{bi+1}\right| + \omega \left\|L_a - L_b\right\|_2 \tag{4}$$

where $|\boldsymbol{M}'_{ai}|$ and $|\boldsymbol{M}'_{bi+1}|$ denote the amplitude of the signal at AoA–ToF a at time i and b at time $i+1$, and ω is the weighting factor. From the above equation, we can see that for the signals reflected from a moving human, their locations change continuously, which makes c_i small. In contrast, dynamic multipath paths change fast and randomly, which makes c_i large. Thus, we can minimize the following total cost function to extract the location of the target:

$$\hat{\boldsymbol{L}} = \arg\min_{\boldsymbol{L}} C = \sum_{i=1}^{T-1} c_i. \tag{5}$$

C. Bounding Box Generation

Dynamic multipath often involves multiple reflections between the human body and the environment. Consequently, it inherently contains exploitable information about both the target and its surroundings. This information can be extracted and leveraged by NNs to assist in bounding box generation for the localized target region. To this end, we need to select a network architecture capable of extracting useful information from multipath reflections while avoiding overfitting to the multipath characteristics of a single environment, which would compromise generalization. By introducing an attention mechanism, the model can focus on the effective region where the target is located. We first identify this effective region using our localization algorithm as a prior step. Through a cross-attention mechanism, we designed a location-mapping cross-attention network to integrate localized region features with dynamic multipath information for simultaneous bounding box generation and fall detection. The sequence of regions detected based on the localization coordinates serves as the query set, while the full heatmap serves as the key-value set. The network architecture is illustrated in Fig. 3. Specifically, given an AoA–ToF map and localized coordinates, we first crop the localized region from the heatmap based on the coordinates. Both the full heatmap and the cropped localized region are then fed into a CNN for feature extraction. After obtaining the localization results for each frame, we crop a 32×32 matrix centered at the localization coordinates to serve as the localized region. We stack the 3-D matrices of seven consecutive frames as the input to the CNN, resulting in seven input channels. A cross-attention module subsequently models the relationship between the localized region features and full heatmap features, enabling the effective utilization of relevant multipath information associated with the target.

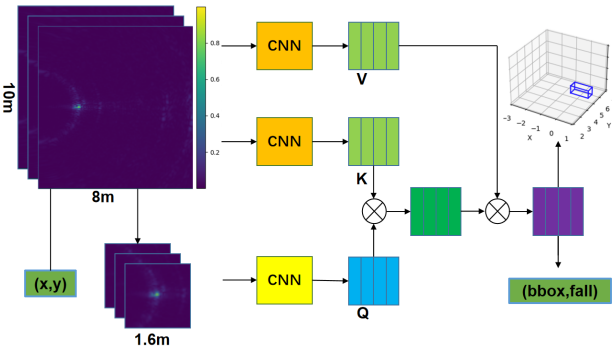


Fig. 3. Overview of the Loc-Map network structure. The features of the entire heatmap serve as the attention mechanism's V and K, while Q is extracted from the features of the localized region.

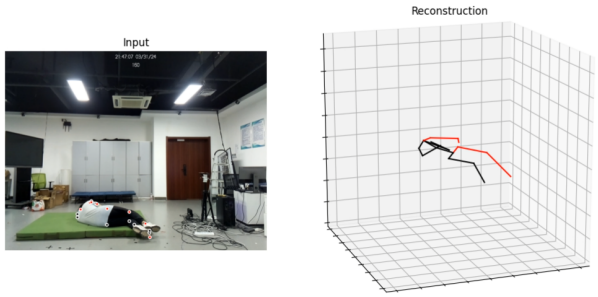


Fig. 4. Ground-truth generation: We generate 2-D keypoints in video sequences using Detectron2 and subsequently generate 3-D skeletal structures through VideoPose3D [45].

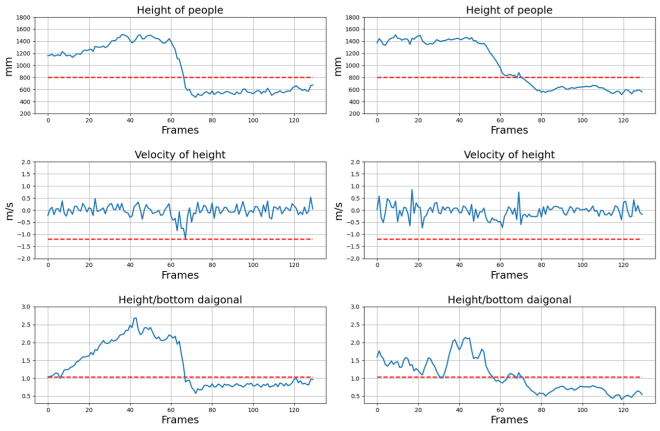


Fig. 5. Two examples for slow falls, these cases exhibit no violent impact yet are likely indicative of anomalous user states. The red dashed line denotes the reference alert threshold.

Crucially, when environmental changes render multipath components as dominant interference, the network dynamically shifts attention toward the localized region features to filter out interference, thereby enhancing system robustness. Finally, a simple fully connected layer aggregates the refined features to produce bounding box coordinates and the fall detection decision.

1) CNN Backbone: We design a lightweight NN to extract fine-grained features, integrating information from radar heatmaps to generate accurate and reliable bounding

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TABLE II
OUR MODEL STRUCTURE

Architecture	Parameters1	Parameters2
Input	[200,160,7]	[32,32,7]
Conv-BN-ReLU	ks=5×5, cs=1, st=1	ks=5×5, cs=1, st=1
Conv-BN-ReLU	ks=5×5, cs=8, st=2	ks=5×5, cs=8, st=1
Conv-BN-ReLU	ks=5×5, cs=16, st=2	ks=5×5, cs=16, st=1
Conv-BN-ReLU	ks=5×5, cs=32, st=2	ks=5×5, cs=32, st=1
Linear-LN-ReLU	64	64

boxes. Our NN comprises a feature encoder and a bounding box predictor. The encoder consists of four convolutional layers, while the predictor consists of three fully connected layers. Due to the specularity of the human body with respect to RF signals, a single RF snapshot can only capture a subset of the human body. Therefore, we stack the heatmaps of the current RF frame with the preceding and subsequent three frames as input for the NN. We first fuse the features of seven frames into one feature map along the temporal dimension using a 2-D convolution. Then, we extract spatial features through a three-layer convolutional encoder. Our NN architecture is illustrated in Table II. The network parameters corresponding to the full heatmap and the localized region are designated as Parameter 1 and Parameter 2, respectively.

2) Cross-Attention Module: Although the backbone network extracts spatio-temporal features from both the full heatmap and the localized region, the heatmap contains environmental information that limits cross-domain generalization, while cropping the localized region discards valuable data, compromising bounding box accuracy. To reconcile these limitations, we introduce a cross-attention module that fuses both information streams. As illustrated in Fig. 3, this module treats the localized region feature vector as the query (Q) and the full heatmap features as key–value pairs (K, V). This process is formulated as follows:

$$\text{Attention}\ (\mathbf{Q}_L, \mathbf{K}_H, \mathbf{V}_H) = \text{Softmax}\left(\frac{\mathbf{Q}_L \mathbf{K}_H^T}{\sqrt{d}}\right)\mathbf{V}_H \tag{6}$$

where d is the dimensionality of the keys, and $\mathbf{Q}_L$ and $\mathbf{K}_H, \mathbf{V}_H$ are the parameters of the query and the key–value pairs, respectively. The module dynamically augments target features with environmental context derived from relevant multipath reflections while suppressing noise by attenuating attention to uncorrelated regions.

3) Loss Function: In order to obtain more precise positions and shapes of bounding boxes, we reference the CIoU loss [42] and 3-D intersection over union (3DIoU) [43] to introduce the 3D-CIoU loss. The 3D-CIoU loss optimizes the normalized distance to ensure fast convergence of the loss function, and strives to make the aspect ratio and position of the predictions as accurate as possible when there is overlap between predictions and targets. The 3D-CIoU loss is composed of three components: 3DIoU, normalized distance, and geometric aspect ratio. The loss function is given by

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2\left(\mathbf{b}, \mathbf{b}^{gt}\right)}{c^2} + v \tag{7}$$

where $\mathbf{b}$ and $\mathbf{b}^{gt}$ denote the central points of the bounding box, $\rho(\bullet)$ is the Euclidean distance, and c is the diagonal length of the smallest enclosing box covering the two boxes. v measures the consistency of the aspect ratio and can be calculated as

$$v = \frac{4}{\pi^2}\left(\arctan\frac{\sqrt{l^{gt2} + w^{gt2}}}{h^{gt}} - \arctan\frac{\sqrt{l^2 + w^2}}{h}\right)^2 \tag{8}$$

where l, w, and h are the predicted length, width, and height, respectively, and l^{gt} and w^{gt}, h^{gt} are the corresponding ground-truth values. Through backpropagation, normalized distance can provide a moving direction for bounding boxes when nonoverlapping with the target box, and the aspect ratio offers clearer geometric features. Optimization can be performed simultaneously on both position and shape, thus obtaining the target's location and state. Binary cross-entropy (BCE) loss is selected as the classification loss. The final loss function for training is

$$L_{final} = \mathcal{L}_{CIoU} + \beta\mathcal{L}_{BCE}. \tag{9}$$

D. Fall Detection and Visualization

Existing fall detection systems only independently output the final fall determination, which hinders users or caregivers from making further informed decisions. To address this, we aim to provide enhanced auxiliary decision-making information while preserving privacy, or to allow for manual correction in case of misjudgment.

During a fall, the height of the 3-D bounding box decreases while the width and length increase. Building upon methodologies from wearable device research [44], we incorporate multithreshold decision logic to enhance fall determination with auxiliary judgment capabilities. We integrate existing wearable methods that detect human velocity with visual methods that assess height. Our approach monitors bounding box height and width across frames. After smoothing the box, the rates of change for height and width are computed via differentiation.

After a fall event, we can assess the severity of the fall in real-time by continuously monitoring the state of the human body using the bounding box. Specifically, when the system detects fall outputs exceeding five instances within the most recent 20-frame window, it enters an alert state and records bounding box heights over subsequent time intervals. If the user's height persists at a low level beyond a predefined time threshold, the system triggers an emergency signal. In our subsequent experiments, this threshold duration is typically set to 10 s.

Furthermore, a gradual fall, where consciousness slowly fades, and the intentional action of lying down to rest share behavioral similarities. Consequently, we can separately detect the abnormal state following a fall event and treat it as an individual alert. Specifically, when a person lies still on the ground for more than 60 s, we can identify this as abnormal behavior and trigger an alert.

IV. EXPERIMENTS

To the best of our knowledge, existing open-source fall datasets are typically limited to a single radar configuration and deployment setup, lacking challenging corner cases that rarely occur in daily life yet critically impact system performance. In this work, we introduce a novel multimodal fall dataset comprising 60 h of radar data synchronized

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with 45 h of video annotations. We subsequently detail the dataset construction methodology, benchmark state-of-the-art approaches on this dataset, demonstrate the efficacy of the proposed location-mapping cross-attention network, and present visualizations of two complex fall scenarios for qualitative analysis.

A. Dataset

Our objective is to establish a comprehensive radar fall detection dataset. To achieve this, we implemented a synchronized acquisition system integrating visual annotation via cameras with dual FMCW radar units: one deployed overhead for top-down vertical coverage and another positioned laterally to expand spatial monitoring capacity.

1) Ground-Truth Annotation: Our annotation schema comprises human bounding box geometry and fall/nonfall classification. To standardize data collection, we synchronized subject behavior with prerecorded audio commands, ensuring specific actions (including falls) were performed within fixed time intervals per data segment, thereby enabling precise action labeling. For bounding box annotations, we leveraged the VideoPose3D [45] framework to extract 2-D skeletal keypoints from multiview video feeds as shown in Fig. 4. These keypoints were then reconstructed into 3-D coordinates using biomechanical constraints, from which minimum bounding cuboids encapsulating all keypoints of each subject were computed to generate supervised training labels for human bounding boxes.

2) Dual-Radar System: We employed two TI MMWCAS-RF-EVM FMCW radar units for synchronized RF data acquisition.

The first radar features a 12-transmitter and 16-receiver MIMO antenna array forming an 86×1 virtual array optimized for high azimuth resolution, configured with single-chirp-per-frame transmission and horizontally deployed at 1–1.5 m height, covering an area of approximately 8×10 m.

The second unit utilizes a three-transmitter and four-receiver MIMO array with an 8×1 virtual array, transmitting 64 chirps per frame to achieve enhanced velocity resolution while being ceiling-mounted at 2.5–3 m height to cover a 4×5 m area. Mutual interference prevention was implemented through nonoverlapping frequency band allocation—77–79 GHz for the first radar and 60–62 GHz for the second. The method proposed in this article utilizes radar data from the first parameter configuration, while the second configuration is employed for dataset augmentation and comparative experiments.

3) Synchronization: The radars and multicamera system were synchronized using the network time protocol (NTP). All devices, including the radars and cameras, were synchronized with a local time server in advance. During data capture, a timestamp for 15 s in the future was sent to the radars and cameras simultaneously via a TCP connection, and all devices remained idle until the designated time. In this way, the time misalignment was primarily caused by the synchronization error between each device and the local time server, which is independent for each device, achieving millisecond-level synchronization accuracy.

Our dataset encompasses 60 h of annotated radar signals, incorporating both vertical and horizontal deployment configurations across five distinct environments with ten diverse viewpoints, collected from over 50 subjects. With an actual acquisition duration of approximately 45 h segmented into per-minute clips, each clip contains 0–4 fall instances, culminating in approximately 6000 annotated falls. The dataset was acquired in two distinct collection phases: an initial 30-h dataset comprising synchronized single-radar and camera-annotated data, followed by two 15-h acquisitions of dual-radar configurations with synchronized video annotation. The detailed description of our dataset is shown in Tables III and IV.

B. Evaluation Metric

The mmFallBbox system integrates two functionalities: bounding box generation and fall detection. For the bounding box generation, we evaluate localization accuracy using IoU with ground truth and height error. For fall detection, performance is assessed using the following metrics.

1) *Accuracy:* The ratio of correctly classified samples over all samples, i.e., $A = TP + TN/TP + TN + FN + FP$. When negative samples outnumber positive samples significantly, $A \approx TN/TN + FP$, which mainly measures the accuracy of negative samples and ignores the accuracy of positive samples, i.e., the falls. Hence, in subsequent tables, accuracy is for reference only.
2) *Precision:* The fraction of correctly detected falls over all detected falls, i.e., $P = TP/TP + FP$.
3) *Recall:* The fraction of correctly detected falls over the total number of falls, i.e., $R = TP/TP + FN$.
4) *$F1$ score:* The harmonic mean of precision and recall, i.e., $F1 = 2 \times P \times R/P + R$.

The dataset was partitioned by environment and user groups to evaluate system performance in cross-environment and cross-user scenarios. Specifically, 5 h of data were allocated for training, with the remaining data used for generalization testing across environments and users. To enable comparison with existing works, we extracted two-second segments from the continuous data. A segment is labeled as a fall if, based on the frame predictions within it, the system would have triggered an alert according to our decision logic: when more than five fall detections occur within a 20-frame window, followed by a sustained low height for 10 s. A total of 589 fall samples were extracted from the 5-h training partition, alongside approximately five times the number of nonfall samples to simulate the sparsity of falls in real-world scenarios.

C. Overall Performance

Since existing fall detection methods do not generate human bounding boxes, quantitative evaluations demonstrate that mmFallBbox achieves IoU scores with the ground truth of 0.913 (same environment, same user), 0.784 (cross-environment, same user), and 0.692 (cross-environment, cross-user), respectively. The corresponding average height errors are 4.2, 7.8, and 9.6 cm, as shown in Table V.

For fall detection, we implemented several benchmarking approaches referencing the core ideas and signal-processing pipelines described in their original publications. It should be noted that, because certain specific implementation details and hyperparameters were omitted in the original texts,

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TABLE III
DETAILED DESCRIPTION OF DATASET1

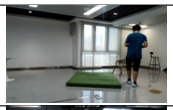
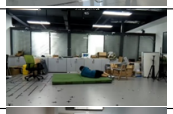
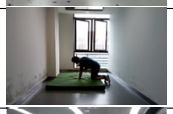
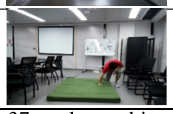
Radar Parameters		Angular resolution: 1.3°, Maximum angle: 120°, Range resolution: 5.7 cm, Maximum range: 8 m		
Patterns of Falls		Falling forward, falling backwards, falling sitting in empty chair and falling sideward		
Daily Activities		Lie down, get up, squat, stand, stand up, sit, walk, run, jump, pick up ,stoop, Push-ups and Sit-ups		
Environments	Environment1 Hall	10 hours, 10 views		Data was captured in a spacious hall environment devoid of clutter or interference sources, with radar signals collected from ten distinct viewpoints for one hour per angle.
	Environment2 Laboratory	10 hours, 10 views		The laboratory environment features a centrally unobstructed area, while the peripheral walls incorporate high-reflectivity objects such as metal cabinets and desks.
	Environment3 Corridor	5 hours, 5 views		The corridor environment is characterized by an elongated, narrow layout with confined lateral space.
	Environment4 Conference Room	5 hours, 5 views		The conference room environment has cluttered arrangements of tables and chairs that induce complex multipath reflections and signal occlusion.
Users		The dataset comprises 37 student subjects (31 male, 6 female), aged 21–30 years, with heights ranging from 160 to 186 cm and body weights spanning 52 to 90 kg.		

TABLE IV
DETAILED DESCRIPTION OF DATASET2

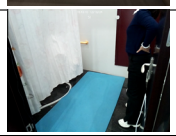
Radar Parameters		Radar1: Angular resolution: 1.3°, Maximum angle: 120°, Range resolution: 5.7 cm, Maximum range: 8 m Radar2: Range resolution: 5.7 cm, Maximum velocity: 1.8m/s , Velocity resolution : 0.056m/s.		
Patterns of Falls		falling forward, falling backwards, falling sitting in empty chair, falling sideward, bed-roll falls , furniture-impact falls, staged-collapse falls, wall-supported falls and post-fall waving gestures		
Daily Activities		lie down, get up, squat, stand, stand up, sit, walk, run, jump, pick up ,stoop, turning in bed, plopping onto sofa, throwing cardboard boxes		
Environments	Environment1 Bedchamber	5 hours, 1 views		Data acquired in the living quarters (bedchamber) focuses on bed interaction scenarios, simulating user-specific behaviors in bedroom environments—including bed-roll falls and controlled bed transfers—while incorporating common interference sources such as footwear removal and rotating fan disturbances.
	Environment2 Livingroom	5 hours, 1 views		Data captured in the living room environment emphasizes user interactions with furnishings, specifically documenting controlled plopping onto sofa, intentional table displacements, incidental furniture contact during falls, and post-impact leaning against furniture objects to simulate realistic domestic accident scenarios.
	Environment3 Toilet	5 hours, 1 views		The toilet scenario data collection encompasses cases of syncope -induced falls during restroom activities and bathing-related falls, incorporating environmental interference such as water flow disturbances and swaying shower curtains.
Users		The dataset comprises 19 volunteers (15 male, 4 female), aged 22–46 years, with heights ranging from 165 to 180 cm and body weights spanning 63 to 85 kg.		

TABLE V
OVERALL PERFORMANCE AND PERFORMANCE OF DIFFERENT PEOPLE AND DIFFERENT ENVIRONMENTS

	3DIOU	Height Error (cm)	Precision	Recall	F1 score	Accuracy
Same Environment, Same User	0.913	4.2	1	0.992	0.996	0.999
Cross Environment, Same User	0.784	7.8	0.985	0.958	0.972	0.998
Cross Environment, Cross User	0.692	9.6	0.972	0.983	0.977	0.996

we determined these missing parameters based on our best understanding of the methods to enable functional evaluations. Therefore, our specific practical implementations might differ slightly from the original authors' exact proprietary systems. We detail the exact framework and hyperparameters we used for each comparative method below,

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and we have made the complete benchmarking source code publicly available at https://github.com/xuanmeikawaii/benchmarking-approaches-of-mmFallBbox.

1) *Point Cloud + Height Drop Baseline (Adapted from TI's Algorithm [46]):* This baseline represents the traditional rule-based point cloud route. We implemented the official processing chain without NNs. After 3-D point cloud extraction and DBSCAN clustering, the maximum z-coordinate was continuously tracked to estimate the target's height. A fall event was triggered according to the standard threshold: a relative height reduction of 60% occurring within a 2.5-s time window (corresponding to a 50-frame buffer at 20 frames per second (fps).
2) *RAM + 3D-CNN Baseline (Adapted from Aryokee [8]):* This baseline represents the multiview spatial heatmap route. Horizontal and vertical RAMs were generated via 2-D beamforming. The input sequence consisted of 112 consecutive frames, resized to 160 $\times$ 200. The network utilized two parallel branches, each containing ten 3D-convolution layers with 5 $\times$ 5 $\times$ 5 kernels and increasing channel dimensions (16, 32, and 64), interleaved with 3-D max-pooling layers. Features from both branches were fused into two fully connected layers. The model was trained using the Adam optimizer (learning rate $=0.002$, batch size $=32$) for 50 epochs.
3) *RDM + ST-ConvLSTM Baseline (Adapted from Ding et al. [35]):* This baseline evaluates the spatiotemporal dynamic feature route. RDMs were extracted using a 0.4-s sliding window with a 50% overlap. For each 2-s interval, nine consecutive RDM frames (resized to 64 $\times$ 64 bins) were fed into the network. The architecture attached a spatial attention module before each ConvLSTM cell (3 $\times$ 3 conv, hidden state dimension of 64). This was followed by a temporal attention module utilizing a shared multilayer perceptron (MLP) with 64 hidden units. The network was trained using the Adam optimizer (learning rate $=0.001$, batch size $=32$) for 50 epochs.
4) *STFT + ResNet Baseline (Adapted from Saeed et al. [31]):* This baseline evaluates the time–frequency analysis route. The FMCW echoes were processed using the STFT with a Hamming window of 256 samples and a 75% overlap to generate continuous micro-Doppler spectrograms. The resulting heatmaps were resized to a standard 200 $\times$ 160 resolution. A standard ResNet-18 architecture, adapted for single-channel input, was utilized as the deep feature extractor. The network was trained via categorical cross-entropy loss using the Adam optimizer (learning rate $=0.001$, batch size $=32$) for 50 epochs.
5) *Point Cloud + Stillness Baseline (Adapted from Zhang et al. [10]):* This baseline evaluates the state-machine point cloud route. 4-D point clouds were extracted with phase compensation. Spatial filtering was performed using DBSCAN with a minimum cluster size of 10 and an epsilon neighborhood radius of 0.2 m. The detection logic relied on a two-stage state machine: an initial rapid height drop detection, followed by postfall micro-motion monitoring, which triggers an alert only if the velocity variance remains below a stillness threshold (0.05 m/s) for three consecutive seconds.

TABLE VI
PERFORMANCE COMPARISON WITH REPRESENTATIVE METHODOLOGICAL BASELINES

	Precision	Recall	F1 score	Accuracy
Point Cloud + Height Drop [46]	0.819	0.787	0.803	0.910
RAM + 3D-CNN [8]	0.914	0.883	0.897	0.975
RDM + ST-ConvLSTM [35]	0.910	0.830	0.871	0.966
STFT + ResNet [31]	0.874	0.815	0.847	0.944
Point Cloud + Stillness [10]	**0.987**	0.823	0.898	0.976
mmFallBbox (Ours)	0.972	**0.983**	**0.977**	**0.996**

TABLE VII
PERFORMANCE COMPARISON OF ABLATION STUDY

		Precision	Recall	F1 score	Accuracy
Full Heatmaps	same env	1	0.994	0.997	0.999
	cross env	0.610	0.521	0.562	0.794
Localized Regions	same env	0.912	0.873	0.896	0.975
	cross env	0.854	0.638	0.768	0.899
Denoising Heatmaps	same env	0.972	0.943	0.957	0.986
	cross env	0.872	0.711	0.783	0.914
Loc-Map Network	same env	1	0.992	0.996	0.999
	cross env	0.972	0.983	0.977	0.996

All experiments were conducted under cross-user and cross-environment conditions, with quantitative results detailed in Table VI.

As shown in Table VI, most baseline methods exhibit suboptimal performance when confronted with certain corner cases. The majority fail to detect slow-motion falls, significantly compromising recall. Meanwhile, the Point Cloud + Stillness baseline [10] demonstrates limited adaptability due to fixed decision thresholds, resulting in misclassifications of bed-fall events or wall-leaning postfall scenarios. By incorporating state-aware mechanisms, our system substantially reduces false alarms while enhancing detection precision.

D. Ablation Study

In this section, we evaluate the functionality of the proposed Loc-Map Network. Comparative experiments were conducted using: 1) full heatmaps individually; 2) localized regions individually; 3) full heatmaps with clutter reduction; and 4) the complete system. Results are presented in Table VII.

As shown in Table VII, while the full heatmap approach achieves comparable performance in identical environments, this occurs because the network overfits to environmental multipath effects on targets, leading to significant performance degradation in different environments. For the localized regions, partial information loss causes marked performance degradation in most environments. However, when environmental conditions change, the removal of interfering signals results in comparatively minor performance degradation. The performance after denoising is slightly higher than that of the cropped region alone, but it does not improve the overall performance compared to using the full heatmap. This is because both region cropping and clutter filtering essentially retain only the signals around the target area while discarding environmental information. However, as a high-velocity action, a fall induces significant clutter energy fluctuations in the environment, which provide valuable information for fall

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TABLE VIII
EVALUATION OF MULTIPERSON SCENARIOS

	Precision	Recall	F1 score	Accuracy
Single Person	0.972	0.983	0.977	0.996
Multi Person	0.974	0.923	0.939	0.983

TABLE IX
EVALUATION OF CORNER CASE

	Sample size	TP or TN	accuracy
Robotic vacuums	100	100	1
Free-falling objects	100	100	1
Fall and Kicking a chair	60	60	1
Fall and Waving arms	60	57	0.95
Rolling off a bed	60	60	1
Falling while showering	60	55	0.917
Fall nearby electric fans	60	60	1

detection. Consequently, clutter suppression does not further enhance the performance of the proposed system.

E. Evaluation of Multiperson Scenarios

In this section, we evaluate the fall detection performance in multiperson scenarios. We first eliminate the trajectory of the initially extracted selected region via masking, and then apply a tracking algorithm to extract the trajectory of the second person by tracking the strongest continuous signal, which is then taken as the selected region. During training, it is only necessary to align the position labels with the localization regions, enabling the network to treat trajectories of other targets as interference, thereby obtaining detection results for each target. In the multitarget detection scenario, one person walked randomly in the room while the other performed the test data collection. Table VIII presents the fall detection performance in multiperson scenarios.

In this scenario, the accuracy is not significantly affected, but the recall rate experiences a 6% decrease. This is because when two targets are too close to each other, the localized region covers both individuals simultaneously, where the height of one person may occlude the second. Furthermore, when the two targets separate, the height of the falling target exhibits an anomalous decrease that does not conform to the characteristics of a typical fall, which impacts performance to some extent.

F. Evaluation of Exceptional Scenarios

In this section, we specifically evaluate corner cases where common fall detection systems are prone to failure. Specifically, we trained the model exclusively on the continuous action dataset collected in the laboratory (Dataset 2) and tested it on the home environment dataset. We identified the locations of these samples based on their labels and evaluated their detection rates.

As shown in Table IX, waving arms or crawling after a fall may have a certain impact on performance. This is because if the struggling duration exceeds 10 s, the system may lose the record of the fall impact and fail to recognize it as a fall, although it can still be detected as an anomaly. The impact of the water flow scenario on performance is more pronounced. Our analysis reveals that flowing water can cause certain localization errors, leading the network to focus on incorrect regions and resulting in misjudgments.

TABLE X
COMPARISON OF PARAMETER AND COMPUTATION COST AGAINST DEEP LEARNING BASELINES

	Parameters	FLOPs	Run time
RAM + 3D-CNN [8]	14M	16G	106.5ms
STFT + ResNet [31]	11.6M	142M	20.7ms
RDM + ST-ConvLSTM [35]	151K	3M	2.3ms
mmFallBbox (Ours)	5.3M	980M	17.1ms

G. Real-Time Detection and Visualization

In this section, we evaluate the computational latency and resource requirements of the system. One of the key strengths of our system is its ability to perform real-time fall detection, even on a commercially available laptop with an Intel i5-8256U CPU and NVIDIA RTX 3060. Specifically, mmFallBbox's processing pipeline comprises three primary stages: beamforming (42 ms), localization algorithms (3.2 ms), and NN prediction (17.1 ms). Since data processing and NN operations execute concurrently, our system achieves real-time detection at up to 20 fps.

We computed the FLOPs and parameter count of our method and compared them with those of other representative NN-based baselines. As shown in Table X, compared with these constructed baselines, our approach achieves significantly improved performance while maintaining real-time detection capability.

We show bounding boxes and threshold curves for two fall samples frequently missed by other systems in Fig. 5. These cases reveal that fall events do not necessarily exhibit high velocity or significant height changes, while prolonged stillness can serve as an effective warning indicator for anomaly detection.

V. CONCLUSION

This work introduces mmFallBbox, an interpretable mmWave radar framework that fundamentally transforms fall detection through continuous 3-D bounding box analytics. By developing the novel Loc-Map architecture—which dynamically utilizes environmental multipath reflections while suppressing interference via a cross-attention network—we overcome critical limitations in cross-environment generalization. Our bounding box-driven paradigm not only enables reliable detection of slow-onset collapses like syncope with a 97.7% $F1$-score but also provides actionable visual explanations through spatiotemporal dimension evolution, allowing caregivers to distinguish high-risk events from benign motions. This framework is validated on a pioneering 60-h RF-Vision dataset featuring vertical/horizontal radar deployments across five environments and 55 subjects under challenging interference conditions.

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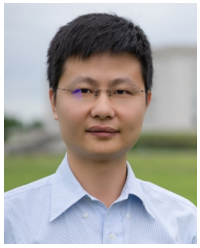
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